Prevalence of Central Obesity in Nigeria; A Systematic Review and Meta-Analysis

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Contents

1 INTRODUCTION **2**

2 METHODOLOGY **3**

2.1 Study Area . . . 3
2.2 Inclusion and Exclusion Criteria . . . 3
2.3 Studies Search Strategies . . . 3
2.4 Data Extraction and Quantitative Analysis . . . 4
2.5 Meta-analytic Model Diagnostics . . . 4
2.6 Analysis of Heterogeneity . . . 4
2.7 Publication Bias Assessment . . . 4

3 RESULTS **4**

3.1 Search Results . . . 4
3.2 Methodological features and the Critical Appraisal of the Included Studies . . . 4
3.3 Characteristics of the Studies Included in the Quantitative Analysis . . . 6
3.4 Fitting the Meta Analytic Models . . . 7
3.5 Models' Diagnostics . . . 7
3.6 Analysis of Between-Study Heterogeneity; Meta-regression using Studies Characteristics as Predictors . . . 8
3.7 Analysis of Publication Bias . . . 8
3.8 Sub-group Analyses of Mean Waist Circumference . . . 8
3.9 Sub-group Analyses of Central Obesity Prevalence . . . 8

4 DISCUSSION **9**

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ABSTRACT
Central Obesity is an excessive accumulation of visceral fat causing an impairment of health. Waist Circumference is a good correlate of abdominal fat and is highly correlated with diabetes, cardiovascular disease and all-cause mortality. This study aims to identify groups with highest prevalence of central obesity in Nigeria. Online searches of Google Scholar, PubMed and Scopus were conducted and studies selected based on predefined criteria. 18 studies consisting of 21859 individuals conducted between 1996 and 2021 were included in the meta-analysis. 39% (95% CI: 25%-54%, I^2 = 99.3%) of adult Nigerians are centrally obese (by IDF criteria). This translates to estimated 46.8 million centrally obese adult Nigerians. More than half [54% (95% CI:39%-68%)] of adult females are centrally obese. This is twice the prevalence in males [13% (95% CI:5%-24%)]. Although southern regions have higher prevalence of central obesity than northern regions in general (28% vs 21%), the northern regions' centrally obese are particularly at a substantially more increased risk for cardiovascular disease (48% vs 18%). The overall mean waist circumference in Nigeria is 83.8cm (95% CI: 80.8cm-86.8cm,I^2 = 99.8%, P* value is <0.001). *Differences in central obesity prevalence rates reflect differences in mean waist circumferences. In conclusion, females, Urban residents and individuals living in the northern part of Nigeria have a high burden of central obesity. Public health measures for stopping the epidemic of obesity should be targeted at these high risk groups.*

Keywords: Obesity, Nigeria, Waist Circumference, Systematic Review, Meta-analysis

1 INTRODUCTION

Obesity is an excessive accumulation of fat in adipose tissue causing an impairment of health[1]. For any given amount of total body fat, there is great variation in the pattern of regional fat distribution with men having about twice the amount of abdominal fat found in pre-menopausal women[2]. Waist Circumference is a good correlate of abdominal fat and thus used to measure central or abdominal obesity i.e. an excessive accumulation of abdominal fat[3]. The Third Report of the National Cholesterol Education Program (NCEP) Expert Panel on Detection, Evaluation, and Treatment of High Blood Cholesterol in Adults (ATP III) defines central obesity as waist circumference of above 102 cm for males and above 88 cm for females[4]. Definitions of central obesity by the The International Diabetes Federation (IDF) are not only gender-specific but ethnicity-specific as well with Europoids cut off values of above 94 cm for males and above 80 cm for females; these values are recommended for African populations until more data are available[5].

A recent meta-analysis of 288 studies conducted worldwide and involving 13,233,675 individuals showed that the prevalence of central obesity is rising from 31.3% in the years 1985–1999 to 48.3% in the years 2010–2014. This increase was found to be higher in males s (from 25.3 to 41.6%) compared with females (from 38.6 to 49.7%). Prevalence of central obesity in Africa was found to be high at 49.7% even though the region is dominated by low and middle income countries[6].

Waist circumference is not only correlated to abdominal fat but also to diabetes, cardiovascular disease and all cause mortality. In a meta-analysis of 50 studies conducted worldwide and involving 2056428 individuals, it was found that a 10 cm increase in waist circumference was associated with 11% higher risk of all cause mortality and this association remained significant and even strengthened after controlling for body mass index[7]. In another meta-analysis of 23 longitudinal observation studies involving 62 study arms with 259,200 individuals, WC above 102 cm in males and above 88 cm in females was a better predictor than BMI above 30kg/m^2 for development of diabetes; this finding was especially strong for women and those above age of 60 years[8].

Despite the importance of central obesity in relation to diabetes, cardiovascular disease and all cause mortality, there are no systematic reviews and meta-analyses of the prevalence of abdominal obesity in Nigeria.

This study aims to estimate the overall, sex-specific and region-specific prevalence rates of central obesity in the country.

2 METHODOLOGY

The systematic review and meta-analysis was conducted according to the Meta-analysis of Observational Studies in Epidemiology (MOOSE) guidelines[9].

2.1 Study Area

Nigeria is a western African nation with an area of 923,769 sq km, home to more than 250 ethnic groups [10]. It has 36 states and a capital divided into 6 geo-political zones or regions; 3 regions each in the northern and southern parts of the country. The estimated population in 2021 is 211.4 million. 43.4% of the population are under the age of 14 years. 53.9% of the population are between the ages of 15 and 64 years. Only 2.8% of the population are above the age of 65 years.[11]. Supplementary material S1 shows the map of the geopolitical zones of the country.

2.2 Inclusion and Exclusion Criteria

Included in this systematic review are population-based studies in Nigeria conducted in individuals aged 15-65 years. Included studies must use either the IDF or NCEP ATP III criteria for diagnosing central obesity. Excluded from this review are studies conducted in pregnant women, studies failing to report prevalence of or the criteria used in defining central obesity and all hospital-based studies.

2.3 Studies Search Strategies

Online databases used were Google Scholar, PubMed and Scopus. Search words and phrases used were: ‘waist circumference,’ ‘central obesity,’ ‘central adiposity’ and ‘abdominal obesity.’Search was repeated for each word or phrase with the name ‘Nigeria.’ Search was conducted between May 2021 to August 2021. Screening of the abstracts and titles of the articles was done independently by two reviewers (AHY and IGM) and studies selection and exclusion done based on the predefined criteria. Thereafter, screening of the full-text articles was done independently by the same researchers to select the studies to be included in qualitative and quantitative analyses.

The methodological quality of the included studies was assessed using a modification of the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Studies Reporting Prevalence Data[12]. The item’s original 9 questions were reduced to six (6) with questions one (1) and two (2) were given a score of two (2) and the remaining four (4) question each given a score of one (1) for a ‘yes’ and zero (0) for a ‘no.’ The total maximum score is eight(8). A study was judged as good quality if it scored minimum of six (6) and of poor quality if it scores less than six (6). Assessment was done independently by two reviewers (AHY and MM) with disagreements sorted by AIH. Supplemental Material S2 shows the modified tool used in critical appraisal of the included studies. The minimum sample size for scoring a study as a ‘yes’ was one hundred and ninety (190) calculated using Epitools[13] online calculator assuming an estimated prevalence of 14.3%, which is the prevalence of generalized obesity in Nigeria based on a recent meta-analysis. Validity of methods was assessed based on whether trained anthropometrists took the waist circumference measurements whereas reliability was assessed based on whether measurements were taken using a consistent landmark (e.g at iliac crest or midpoint between rib cage and iliac crest) . Only studies judged as having good methodological quality were included in the quantitative analysis.

2.4 Data Extraction and Quantitative Analysis

Data extraction was independently done by two reviewers (MAB and IGM). Extracted information from the studies included prevalence of central obesity, mean and standard deviation of the waist circumference, sample size, settlement (urban/rural), state and region of the study, study year, mean age and sex composition of the study participants. Data was entered into Excel and then imported into R statistical environment for statistical computing, version 4.1.0.1[14]. Metafor Package[15] was used to fit Random Effects Model for pooling prevalences and Mixed Effects Model for meta-regression using inverse variance method with correction of pooled estimate and its variance using Sidik-Jonkman's estimator for between study heterogeneity[16].

2.5 Meta-analytic Model Diagnostics

The diagnostics of the meta-analytical model was carried out according to methods previously described[17]. Briefly, Individual studies were investigated for outliers and influencers based on hat value (standardized distance of each study's reported prevalence from the pooled prevalence), rstudent (standardized distance of the predicted prevalence for each study from the pooled prevalence), Cook's distance (the distance between the pooled prevalence when the individual study is included and when it is excluded) and diffits (distance between pooled prevalence with the study included and when the study is excluded but in standard deviation units). Cut off values implemented in the R metafor package were adopted[15].

2.6 Analysis of Heterogeneity

Meta-regression model, using characteristics of the included studies as predictors, was fitted to investigate the extent to which the calculated heterogeneity is attributed to the study-level characteristics such as the sex composition, geo-political region, settlement, mean age of the subjects and sample size. Pseudo-R^2, defined as percentage of explained heterogeneity after fitting the meta-regression model, would be used to estimated the amount of heterogeneity due to study level characteristics.
Prediction Intervals were reported to overcome the difficulties in interpreting both tau^2 and I^2 as measures of between study heterogeneity[18]

2.7 Publication Bias Assessment

Funnel plot was used to visually inspect for possible publication bias where studies reporting small prevalence were not published and thus not included in the meta-analysis. Formal regression test develooped by Egger and colleagues[19] was employed for testing funnel plot asymmetry.

3 RESULTS

3.1 Search Results

Figure 1 shows the results of the search strategy.

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3.2 Methodological features and the Critical Appraisal of the Included Studies

40 studies were selected and included in the qualitative analysis of methodological features. The quality appraisal of the included studies is shown in the supplemental material S3.

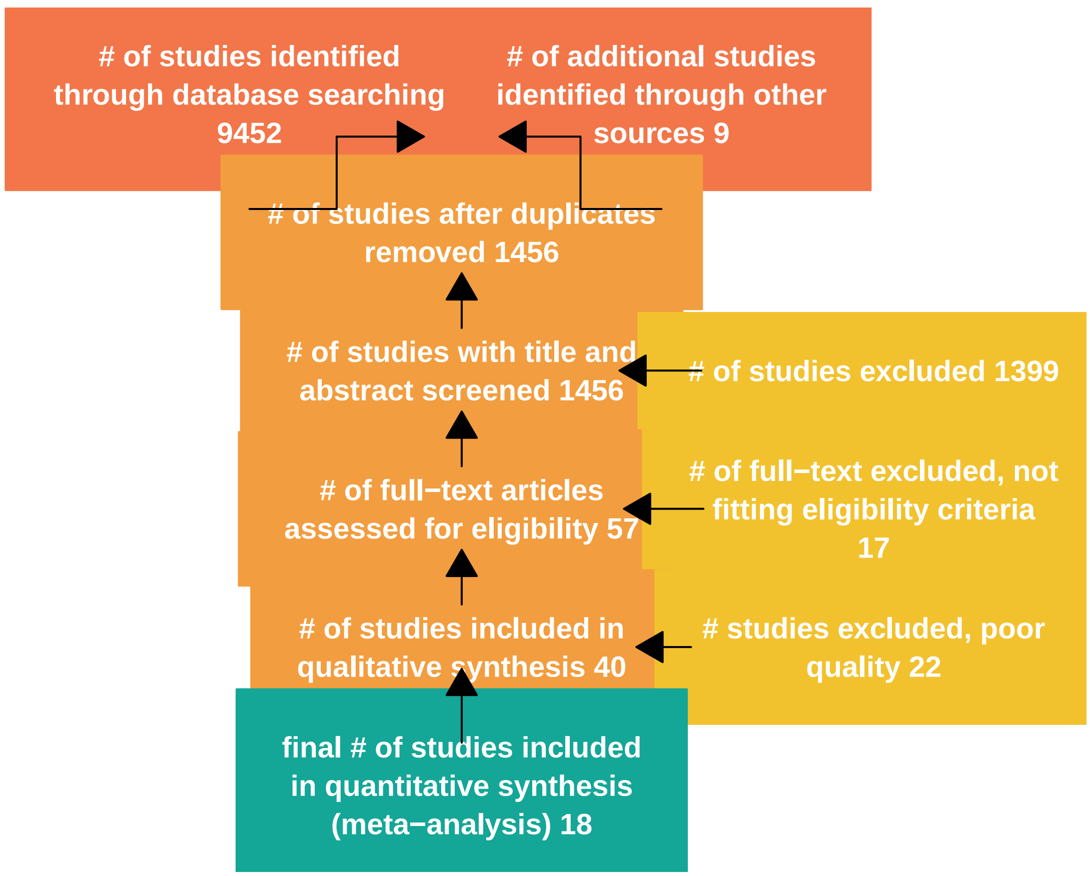

Figure 1: Results of the Search

3.3 Characteristics of the Studies Included in the Quantitative Analysis

18 studies consisting of 21859 individuals met the inclusion criteria and are of good or moderate quality and thus included in the meta-analysis. The studies were conducted between 1996 and 2021. There were 2 studies from the north-west region; 1(one) conducted in an urban settlement and the other conducted in both rural and urban settlements. Similarly, there were 2 studies from the north_central 1(one) conducted in an urban settlement and the other conducted in both rural and urban settlements. There were 6 studies from the south-east with 1 study conducted in an urban, 2 studies in a rural and 3 studies in both rural and urban settlements. There is only 1 study conducted in the south-south region.Similarly, there is only 1 study conducted in the north-east region . 6 studies were conducted in the south-west region including 4 studies in urban areas, 1 study in a rural area and 1 study conducted in both urban and rural areas. Overall, eight(8) studies were conducted in the urban settlements only, three(3) studies were conducted in rural settlements only and 7 studies were conducted in both rural and urban settlements (Figure 2).

11 of the studies used IDF criteria in defining central obesity whereas 7 studies used ATP III criteria. The region with the youngest study participants was the north-central with reported studies' mean age of 31. South-east studies have the oldest participants with mean age of 48. In general, studies conducted in the southern regions included older subjects than those included in northern regions. The largest studies were from south-west region with mean sample size of 1872. South-south studies are smallest with mean age of 422 (Figure 2)

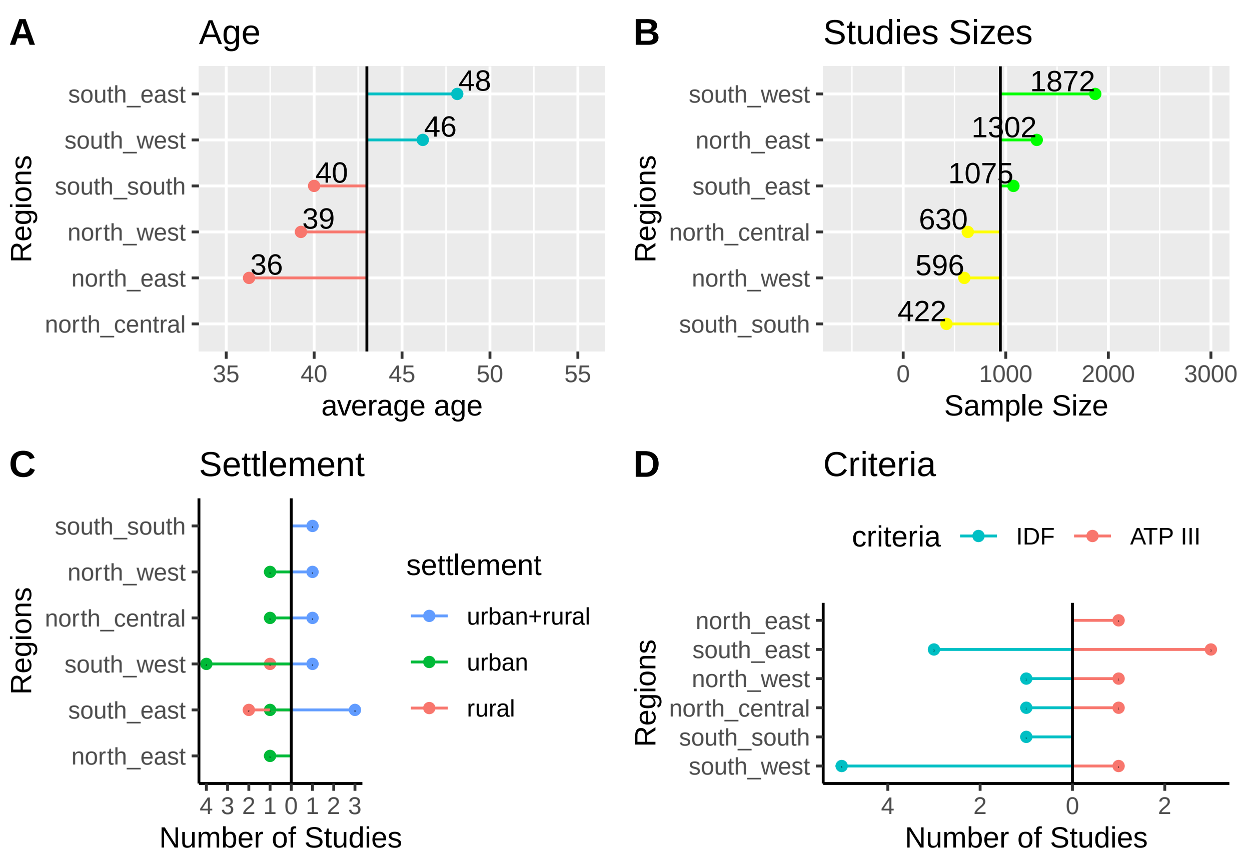

Figure 2: Characteristics of the Included Studies

3.4 Fitting the Meta Analytic Models

Random Effects Model was fitted using inverse variance method with correction of pooled estimate and its variance using Sidik-Jonkman's estimator for between study heterogeneity. Prevalence rates were tranformed using arcsine transformation and means were log transformed. Figure 3 shows the forest plot of the models. The pooled prevalence of central obesity under the IDF criteria,39% (95% CI: 25%-54%), is higher than the value of 24% (95% CI: 12%-38%) under the ATP III criteria. The P values for the random meta analytic model is 0.141, the estimated total between studies heterogeneity not explained by sampling error (I^2) is 99%. Test for Heterogeneity is significant with p-value<0.001 indicating substantial heterogeneity between the included studies. The prediction interval of the pooled prevalence is 2.2%-73.9%.

The overall mean waist circumference in Nigeria is 83.8cm (95% CI: 80.8cm-86.8cm). The P values for the pooled mean model is <0.001, the estimated total between studies heterogeneity not explained by sampling error (I^2) is 99.8%. Test for Heterogeneity is significant with p-value<0.001 indicating substantial heterogeneity between the included studies. The prediction interval of the pooled mean waist circumference is 72.9cm-96.2cm.

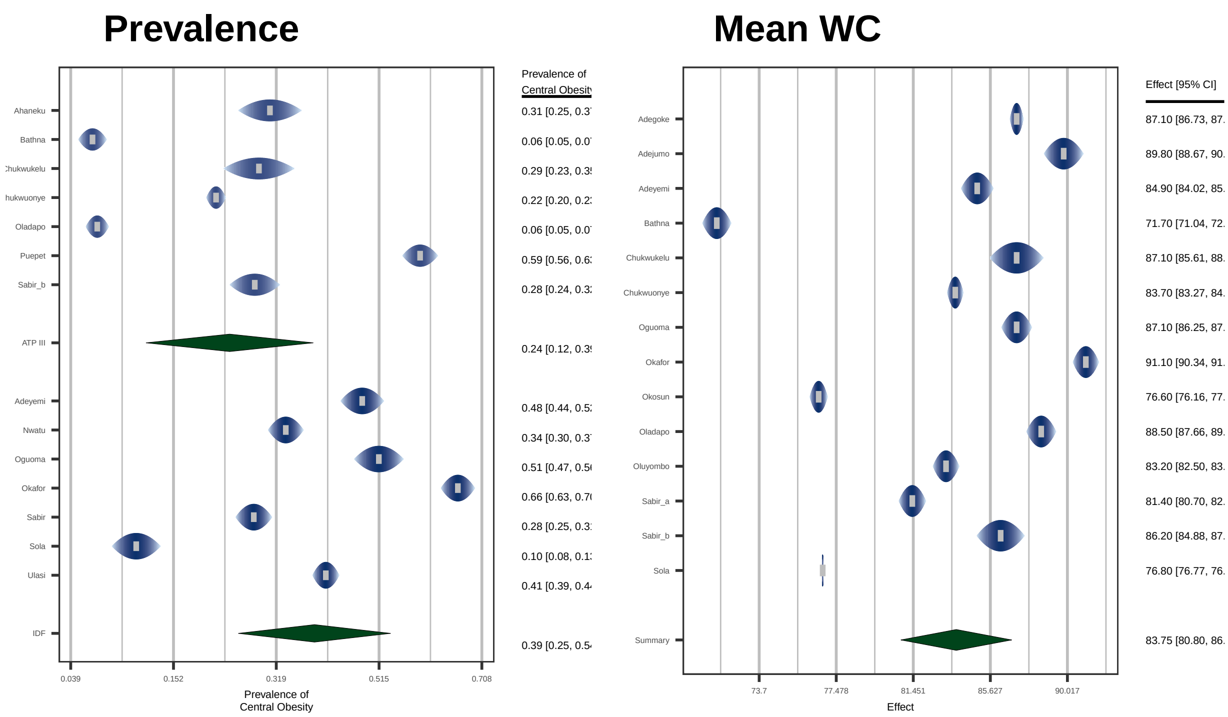

Figure 3: forest plots of Pooled Central Obesity and Mean Waist Circumference

3.5 Models' Diagnostics

Supplemental Materials S4 and S5 showed the graphical output of the influence analysis of the pooled central obesity prevalence and mean waist circumference models respectively. No study was tagged influential in both models.

3.6 Analysis of Between-Study Heterogeneity; Meta-regression using Studies Characteristics as Predictors

A meta-regression model was fitted using geo-political region, settlement, criteria and participants' mean age of the included studies. For the prevalence of central obesity prevalence model, value of pseudo-R^2 after fitting the meta-regression model was 87.3%. The corresponding pseudo-R^2 for the mean waist circumference was 82%. This means most of the heterogeneity between the studies results from the differences in study characteristics. The relative changes in the two heterogeneity measures (I^2 and tau^2) after fitting the meta-regression models for both pooled prevalence of central obesity and pooled mean waist circumference are shown in Supplemental Material S6. Supplemental Material S7 shows the model coefficients of the meta-regression models.

3.7 Analysis of Publication Bias

The funnel plots of the pooled prevalence and mean waist circumfernce models are shown is Supplemental Material S8. There was no obvious asymmetry in the plots. Formal test for plot asymmetery (regression test) was conducted and it was not statistically significant (P value is 0.597 and 0.114 for the pooled prevalence rate and mean waist circumference respectively), confirming the visual assessment of the funnel plot.

3.8 Sub-group Analyses of Mean Waist Circumference

As shown in plot A of Figure 4, the mean waist circumference in the urban settlement, 84cm (95% CI: 80cm-88cm) is higher than the value of 82cm (95% CI: 77cm-88cm) in rural settlements. The mean circumference of females,83cm (95% CI: 80cm-86cm) is very similar to the value of 82cm (95% CI: 79cm-86cm) in males(plot B of Figure 4). The mean waist circumference in the southern regions, 86cm (95% CI: 83cm-89cm) is higher than the northern regions' value of 79cm (95% CI: 74cm-83cm) (plot C of Figure 4). The P values for differences between regions, sexes and settlements are 0.014, 0.806 and 0.727 respectively.

3.9 Sub-group Analyses of Central Obesity Prevalence

As shown in Table 1, Estimates of prevalence under the IDF criteria are consistently higher than estimates under the ATP III criteria. Additionally, females, Urban residents and individuals from the southern regions have consistently higher prevalence rates of central obesity under both ATP III and IDF criteria than their Males, Rural and Northern region residents counterparts. The exception is the prevalence of central obesity in the Northern regions under the ATP III which was found to be higher than the corresponding estimate in the southern regions.

Table 1: Prevalence Rates by Gender, Area of Residence and Region

Groups	X	Prevalence..95..CI.	X.1	I.squared	Ps
	ATP III		IDF		
Gender					
Females	0.359(0.210-0.523)		0.536(0.392-0.677)	98.6	
Males	0.085(0.017-0.200)		0.125(0.046-0.235)		
Residence					

Groups	X	Prevalence..95..CI.	X.1	I.squared	Ps
Rural	0.148(0.021-0.361)		0.279(0.126-0.464)	99.1	
Urban	0.253(0.094-0.457)		0.449(0.280-0.265)		
Regions					
Northern	0.282(0.114-0.490)		0.183(0.031-0.422)	99.4	
Southern	0.206(0.078-0.374)		0.482(0.318-0.647)		

4 DISCUSSION

The higher pooled crude prevalence of central obesity under the IDF criteria (39%) compared to pooled prevalence under the ATP III criteria (24%) is due to the fact that the former criteria uses a lower waist circumference threshold than the latter resulting in higher number of individuals diagnosed with obesity[4,20]. According to the latest data by the United Nations[11], this translates to estimated 46.8 million and 28.5 centrally obese individuals in Nigeria under the IDF and ATP III criteria respectively. The two thresholds have been recognized by WHO to represent different levels of cardiovascular risk with the former denoting increased risk and the latter substantially increased risk, at least in Europoid populations[1]. It has also been suggested that the two criteria correspond respectively to BMI levels of overweight (>/= 25kg/m^2) and obesity (>/= 30kg/m^2)[21]. The pooled prevalence under the IDF criteria is thus a reflection of prevalence of overweight and obesity. A recent meta-analysis conducted in Nigeria found the combined prevalence of overweight and obesity to be 39.3%[22], which is very close to the prevalence of central obesity under IDF criteria we found in this meta-analysis (39%). The prevalence of central obesity in Nigeria is lower than the global average (41.5%) and the average in the African continent (45.7%) but similar to the average in middle income countries, based on a recent large meta-analysis [6].

Analysis of heterogeneity in this analysis reveals different behaviours for the two measures of heterogeneity; I^2 and tau^2. The former is known to be sensitive to the size of studies i.e. if the included studies are fairly large then the sampling error will be close to zero and I^2 as a ratio will approach 100%[23]. To measure, with a precision of 0.05 and 95% confidence interval, a central obesity prevalence of 39% found in this study(under IDF criteria), the minimum sample size required is Three Hundred and Sixty Six [13]. The studies included in this meta-analysis are, by the selection criteria, relatively large with sample sizes having an interquartile range of 959. Consequently, I^2 might be closer to 100% even if the between study heterogeneity is not substantial. This is likely the case as a metaregression reduced the tau^2 by almost 87%. However, I^2 barely changed (Supplementary Material S6). The prediction intervals for the pooled prevalence of central obesity (2%-74%) means the possible values members of a population of highly heterogeneous studies conducted or to be conducted in Nigeria could find.

The higher prevalence of central obesity among females compared to males reflects the fact that females and males have similar mean waist circumference. Considering the fact that, central obesity in females is defined at a lower waist circumference value under both ATP III and IDF criteria, similar mean waist circumference in the two sexes will mean higher prevalence rates of obesity among females. Similar higher prevalence in females was found in measure of generalized obesity (BMI) in Nigeria[22]. This is also a trend observed globally [6]. In Nigeria, this may be due to the fact that women are less active than men as found in a recent meta-analysis [24]. This sexual dimorphism in central obesity was also known to be underlined by biological factors with the male hormone,testosterone, known to favor greater central fat deposition[25].

The higher prevalence of central obesity in the urban settlements compared to rural areas reflects the higher mean waist cicumference in the former. The reason is due to the fact that urban residents are less physically active due to access to motorized transport, clerical and mechanized work activity and access to work-saving

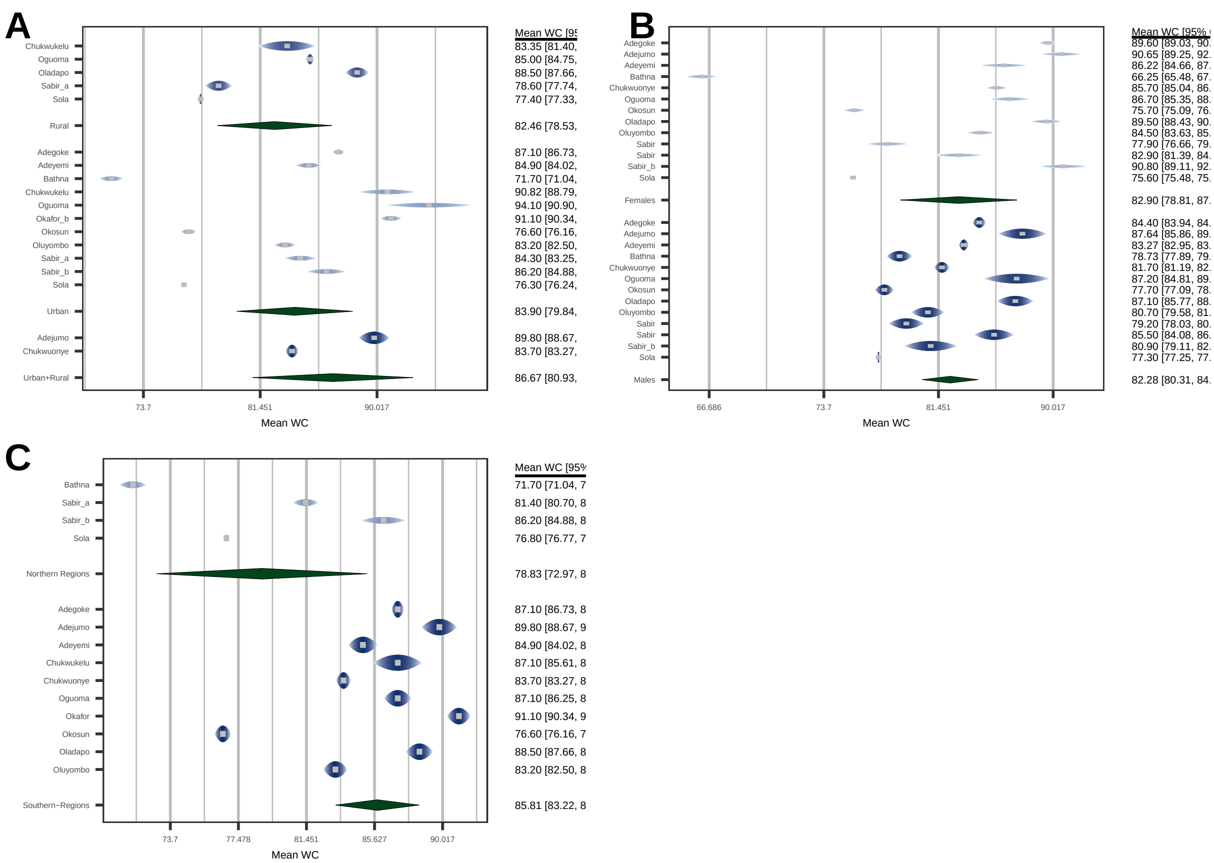

Figure 4: Mean Waist Circumference by Settlement (A), Gender (B) and Region(C)

devices even for home related activities like cooking and laundry. Access to energy dense foods like refined sugar and saturated fats is also higher in urban settlements [1].

The differences between northern and southern regions are more complex. Estimates of IDF criteria, reflecting combination of individuals with increased and those with substantially risk for cardiovascular disease, reveal higher prevalence in the southern compared to northern regions. This is due to the fact that southern regions have higher levels of socio-economic development and rates of urbanization [26,27]. However, if ATP III criteria is used, measuring proportion of those at substantially increased risk for cardiovascular disease only, the northern regions have a higher prevalence. This means although southern regions have higher prevalence of central obesity in general, the northern regions' centrally obese are particularly at a substantially more increased risk for cardiovascular disease.

5 SUMMARY AND CONCLUSION

39% of adult Nigerians are centrally obese (by IDF criteria). This translates to estimated 46.8 million centrally obese adult Nigerians. More than half [54% (95% CI:39%-68%)] of adult females are centrally obese. This is twice the prevalence in males [13% (95% CI:5%-24%)]. Although southern regions have higher prevalence of central obesity than northern regions in general (28% vs 21%), the northern regions' centrally obese are particularly at a substantially more increased risk for cardiovascular disease (48% vs 18%). Differences in central obesity prevalence rates reflect differences in mean waist circumferences.

Females, Urban residents and individuals living in the northern part of Nigeria have a high burden of central obesity. Public health measures for stopping the epidemic of obesity should be targeted at these high risk groups.

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